

Topic 2: Atomic Structure and the Periodic Table

Students will be assessed on their ability to:

2.1	know the structure of an atom in terms of electrons, protons and neutrons
2.2	know the relative mass and charge of protons, neutrons and electrons
2.3	know what is meant by the terms 'atomic (proton) number' and 'mass number'
2.4	be able to use the atomic number and the mass number to determine the number of each type of subatomic particle in an atom or ion
2.5	understand the term 'isotope'
2.6	understand the basic principles of a mass spectrometer and be able to analyse and interpret mass spectra to: - i deduce the isotopic composition of a sample of an element - ii calculate the relative atomic mass of an element from relative abundances of isotopes and vice versa - iii determine the relative molecular mass of a molecule, and hence identify molecules in a sample - iv understand that ions in a mass spectrometer may have a 2+ charge
2.7	be able to predict mass spectra, including relative peak heights, for diatomic molecules, including chlorine, given the isotopic abundances
2.8	be able to define first, second and third ionisation energies and understand that all ionisation energies are endothermic
2.9	know that an orbital is a region within an atom that can hold up to two electrons with opposite spins
2.10	understand how ionisation energies are influenced by the number of protons in the nucleus, the electron shielding and the sub-shell from which the electron is removed
2.11	know that ideas about electronic structure developed from: - i an understanding that successive ionisation energies provide evidence for the existence of quantum shells and the group to which the element belongs - ii an understanding that the first ionisation energy of successive elements provides evidence for electron sub-shells
2.12	be able to describe the shapes of <i>s</i> and <i>p</i> orbitals
2.13	know that orbitals in sub-shells: - i each take a single electron before pairing up - ii pair up with two electrons of opposite spin
2.14	be able to predict the electronic configuration of atoms of the elements from hydrogen to krypton inclusive and their ions, using <i>s</i> , <i>p</i> , <i>d</i> notation and electron-in-boxes notation
2.15	understand that electronic configuration determines the chemical properties of an element

2.17	be able to represent data, in a graphical form (including the use of logarithms of first ionisation energies on a graph) for elements 1 to 36 and hence explain the meaning of the term 'periodic property'
2.18	be able to explain: - i the trends in melting and boiling temperatures of the elements of Periods 2 and 3 of the Periodic Table in terms of the structure of the element and the bonding between its atoms or molecules - ii the general increase and the specific trends in ionisation energy of the elements across Periods 2 and 3 of the Periodic Table - iii the decrease in first ionisation energy down a group